

When Do Zero-Shot Foundation Features Matter for Cross-Session Object Identity?

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Abstract—Open-vocabulary object mapping systems assume that cross-session object identity can be maintained by the zero-shot features they already carry. We test that assumption on the official 3RScan validation split under a single open-set protocol, under oracle instance masks and known scan alignment. Under known alignment a single centroid distance reaches 84.8 open-set F1, ahead of the strongest frozen global embedding (DINOv3, 67.8) and of the strongest appearance condition of any tier (DINOv3 dense patches with an OASIS-style confidence heuristic, 72.4); fusing appearance with geometry adds only 3.1 points, and on same-class-distractor axes appearance never overtakes geometry at all. The ranking inverts once the centroid cue itself becomes unreliable. Beyond one metre of *measured* aligned-centroid distance, appearance reaches 59.5 F1 against the centroid prior’s 9.7, and that regime is $\sim 81\%$ static objects whose centroids only *measure* far apart, consistent with view-dependent partial-view truncation. Perturbing either oracle behind the spatial prior, the known alignment or the mask-given centroids, erodes its advantage along nearly coincident curves against injected centroid error and inverts the ranking near 0.30 m of injected RMS. Over the two noise structures tested, centroid error is a sufficient statistic for that inversion. A lifelong semantic map can therefore spend its GPU budget where the geometric cue is untrustworthy: escalating only where the geometric matcher abstains beats the backbone-free prior by 1.99 F1 while avoiding 81.1% of rescan-side backbone passes, staying within 1.33 F1 of always-on fusion.

Index Terms—lifelong mapping, object re-identification, foundation models, open-vocabulary mapping, semantic SLAM

I. INTRODUCTION

Object-level SLAM and 3D scene graphs have matured [1], [2], [3], spatio-temporal mapping has been formalized [4], and open-vocabulary mapping is now a fast-growing family of systems [5], [6], [7], [8]. They share an implicit assumption: that object identity across mapping sessions, “is this the same chair I mapped last week”, is solvable by the very zero-shot features the systems already carry. A lifelong semantic map stands or falls on that step, and it has not been measured under a controlled protocol.

The available evidence is fragmented. Trained 3D descriptors are evaluated closed-set on the official split (Living Scenes [9]). A trained 4D instance segmentation benchmark reports temporal identity as a training problem [10]. A patch-level system built on frozen features is evaluated on a partially re-annotated 3RScan [11] sequence of 36 annotated objects, 23 of them changed, plus a custom car-park sequence (OASIS-Map [12]). Deployed persistence systems couple CLIP with geometric overlap and ICP [13]. 2D re-identification studies are category-gated [14], scored within a single video [15], or enrol each instance from a few annotated shots [16]. Clos-

est in setting are a frozen-DINOv3 re-identification memory for indoor revisits that is deliberately pose- and depth-free, and so carries no geometric prior to compare against [17], and trained scene-graph alignment that fuses vision-language with geometric features without a geometry-only control [18]. Controlled probes of frozen features exist for other geometric capabilities [19], but none that we are aware of targets cross-session instance identity, open-set, against a backbone-free geometric baseline.

We run the comparison on the official 3RScan validation split [11]: spatial priors, frozen global instance embeddings, set-to-set matching of per-view embeddings, and dense patch correspondence, all open-set and all under one protocol. Every condition is evaluated under oracle instance masks and known scan alignment, which isolates representation capability from detection, segmentation, and registration error.

Three results follow. First, on the pooled validation split geometry wins: under known alignment a backbone-free centroid distance beats every frozen embedding we tested, by 17 F1 points over the strongest global one. Second, the ranking inverts in a specific regime. Where the measured centroid distance between corresponded instances is large, appearance becomes the better cue. Four in five of the objects in that regime never moved, and their centroids are far apart only as measured, under partial-view truncation. Third, the geometric advantage rests on two oracles, and both act through one scalar. Perturbing the alignment and perturbing the mask-given centroids trace nearly the same degradation curve against injected centroid error and invert the ranking near 0.30 m. Escalating to a GPU backbone can therefore be conditioned on centroid trustworthiness, and we give the two numbers such a policy needs.

II. PROTOCOL

Data. 3RScan [11] provides real cross-session rescans with ground-truth instance correspondences, rigid transforms for moved instances, and ambiguity annotations. We use the official train/validation split. Every threshold and hyperparameter is tuned on train and the validation split is evaluated once. Validation contains 110 (reference, rescan) pairs over 47 reference scenes and 2844 both-present correspondence pairs.

Conditions. Every condition runs under oracle instance masks and known scan alignment (the single raw-bounding-box probe of Sec. III-C excepted): instance masks come from the ground-truth segmentation and the cross-session transform is the annotated one. The mask oracle matches the input protocol of the trained-3D reference’s main results [9].

Section III-C then perturbs both oracles and measures what the ranking depends on.

Representations. Per instance we take the $K=5$ views with the largest visible mask area, spaced apart in the sequence, crop to the mask with a small buffer, and extract frozen features from DINOv2, DINOv3, CLIP, and SigLIP [20], [21], [22], [23] at each backbone’s native resolution. Per-view vectors are L_2 -normalized before aggregation; the DINOv3 vector is the CLS token. Three tiers span the space between a pooled vector and dense correspondence: *global* mean-aggregated embeddings, *set-to-set* matching that keeps the K per-view vectors unpooled, and *dense patch* correspondence, scored either by a bidirectional mutual-nearest-neighbour vote or, following OASIS-Map [12], by additionally rejecting pairs whose patch match rate is high while their mean match confidence is low, the signature of two instances of the same category. The spatial prior is the centroid distance after applying the known alignment; the combined score is $\alpha \cos + (1 - \alpha)$ spatial with α tuned on train. Crop construction, view budget, and assignment policy are held fixed across all rows, so a row differs from another only in representation.

Matching and metric. Candidates are all instance pairs within a (reference, rescan) pair: no spatial gate prunes the search, so a far-apart true match is reachable by every condition. Matching is mutual nearest neighbour with a rejection threshold, so every condition may decline to match, and open-set precision, recall, and F1 are reported micro-pooled. True positives are counted as a maximum one-to-one matching [24] on the bipartite graph of allowed prediction/ground-truth edges, where transitively closed ambiguity groups define allowance; counting per edge instead admits double counting inside an ambiguity group. Intervals in Table I are a pair-level percentile bootstrap ($B=1000$) over the 110 pairs; the headline gap is cross-checked below with a reference-clustered BCa bootstrap ($B=4000$), since rescans sharing a reference scan are not independent.

III. RESULTS

A. A centroid beats every frozen embedding on average

Table I gives the comparison. Under known alignment the spatial prior reaches 84.8 open-set F1 (90.8 P / 79.5 R) while the strongest frozen global embedding, DINOv3, reaches 67.8. A paired reference-clustered bootstrap, with both arms restricted to the instances every condition can represent, puts that gap at -16.7 F1 for appearance, 95% CI $[-19.6, -13.5]$, $p < 0.001$; recomputing the two marginal intervals as reference-clustered BCa ($B=4000$ over the 47 reference scenes) widens them to $[80.1, 87.6]$ and $[62.0, 71.9]$ and leaves them non-overlapping as well. Fusing appearance with geometry is best overall at 87.9 (DINOv2 ViT-B + spatial). The DINOv3 fusion ties it at 87.8, and the stratified analyses below use DINOv3 so the fused and unfused appearance conditions share a backbone. That is 3.1 points over the backbone-free geometric prior on the pooled set, or $+3.4$ paired, 95% CI $[+2.2, +4.7]$, $p < 0.001$, a real but small return. The spread

TABLE I
OPEN-SET CROSS-SESSION INSTANCE MATCHING ON THE OFFICIAL 3RSCAN VALIDATION SPLIT, UNDER ORACLE INSTANCE MASKS AND KNOWN SCAN ALIGNMENT. MICRO-POOLED F1, PAIR-LEVEL PERCENTILE-BOOTSTRAP 95% CIs ($B=1000$, 110 (REFERENCE, RESCAN) PAIRS). COST IS ONE BACKBONE PASS, BATCH 1, RTX 4090; INSTANCES AGGREGATE $K=5$ VIEWS. BOLD: BEST OVERALL, BEST BACKBONE-FREE. THE SPATIAL PRIOR IS HANDED THE ANNOTATED CROSS-SESSION TRANSFORM, COMPUTED OUTSIDE THIS ACCOUNTING.

Condition	F1	95% CI	ms/view	GFLOPs
<i>Spatial prior (no backbone pass)</i>				
Centroid + threshold	84.8	82.5–86.9	—	—
Centroid + volume ratio	84.0	82.1–85.8	—	—
<i>Frozen global embeddings</i>				
DINOv2 ViT-B/14	64.2	60.5–67.7	47.0	22.0
DINOv2 ViT-L/14	63.7	60.1–67.0	90.8	77.8
DINOv3 ViT-B	67.8	64.4–70.8	62.7	17.2
CLIP ViT-B/16	49.1	45.1–52.7	42.8	16.9
CLIP ViT-L/14	53.1	49.6–56.4	82.6	77.8
SigLIP base	58.2	54.5–61.6	49.2	16.9
SigLIP large	59.9	56.0–63.4	91.2	175.3
<i>Unpooled and dense (DINOv3 ViT-B)</i>				
Set-to-set over K views	60.3	56.9–63.2	62.7	17.2
Dense patch, MNN vote	67.0	63.3–70.1	62.7	17.2
Dense patch, OASIS heuristic	72.4	69.1–75.5	62.7	17.2
<i>Fusion, $\alpha \cos + (1 - \alpha)$ spatial</i>				
DINOv2 ViT-B + spatial	87.9	85.9–89.7	47.0	22.0
DINOv3 ViT-B + spatial	87.8	85.8–89.6	62.7	17.2

over all seven fused variants is far inside any single interval; 87.9 is the largest.

The fusion gain is close to feature-independent. An 18.7-point spread across feature-only conditions (CLIP-B 49.1 to DINOv3-B 67.8) compresses to a 1.6-point spread once each is fused with geometry (combined CLIP-B 86.3 to combined DINOv2-B 87.9): with geometry alone at 84.8, the choice of frozen backbone barely moves the fused operating point. Among the frozen embeddings both DINO backbones lead both image-text contrastive ones, and the marginal intervals of Table I separate every DINO-vs-CLIP pair and DINOv3 from SigLIP; the two adjacent pairs DINOv3–DINOv2 and DINOv2–SigLIP overlap, so we read the family gap and leave the ordering open.

Neither unpooled nor dense matching closes the gap. Keeping the K per-view embeddings and matching set-to-set falls below pooled aggregation under a maximum-cosine statistic (60.3 vs. 67.8) and recovers only to roughly its level under a chamfer statistic. A plain dense patch mutual-nearest-neighbour vote is at or below global aggregation for both DINO backbones. Only the OASIS-style confidence heuristic [12] adds a modest but clear gain, taking DINOv3 dense patches to 72.4, the strongest appearance condition of any tier and still 12.4 points under the centroid. The deficit is therefore a property of these representations rather than of the pooling step that reads them.

B. The ranking inverts where the centroid cue breaks

The averages hide a sharp regime structure. Fig. 1 traces performance against the measured aligned-centroid distance between corresponded instances, which is the quantity the spatial prior itself consumes; bins are computed over the 2844 both-present correspondence pairs, so the axis exists only

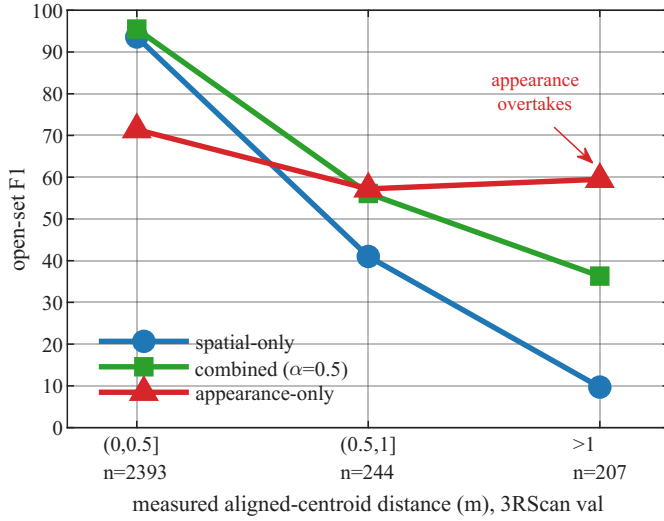


Fig. 1. Open-set F1 against *measured* aligned-centroid distance between corresponded instances, over the 2844 both-present correspondence pairs of 3RScan validation. Spatial-only collapses as this distance grows; appearance-only (DINOv3 global) barely reacts, and passes both the prior and the fixed combination inside the (0.5, 1] bin, with the gap widening beyond a metre. The far bin is $\sim 81\%$ ground-truth-static objects (per-bin static/moved/non-rigid 2046/301/46, 195/43/6, 167/39/1).

where a correspondence does. The centroid prior degrades steeply along that axis, from 93.6 F1 below half a metre to 41.0 in (0.5, 1] and 9.7 beyond a metre. Because the prior uses a train-tuned metric rejection threshold, the far-bin figure reflects the configured matcher as well as the cue itself. Appearance (DINOv3 global) barely reacts along the same axis (71.4 / 57.2 / 59.5), so the appearance-minus-centroid gap rises from -22.2 in the smallest bin through $+16.2$ in the middle one to $+49.8$ beyond a metre. The contrast between the extreme bins’ gaps is $+72.0$ points, 95% CI $[+59.9, +80.0]$ on a shared reference-clustered resample. Only “beyond one metre: appearance > centroid” was pre-registered as confirmatory; the remaining strata are exploratory under Benjamini–Hochberg control.

How far out the crossing sits depends on how much geometry the prior uses. A centroid-plus-volume-ratio prior, overall inside the plain centroid’s interval (84.0 vs. 84.8), is markedly more robust along this axis: 63.6 in (0.5, 1] and 18.0 beyond a metre, against 41.0 and 9.7. It therefore holds off appearance through the middle bin (63.6 vs. 57.2) and yields only beyond a metre (18.0 vs. 59.5). Appearance overtaking geometry beyond a metre survives a stronger geometric prior; where exactly the two curves cross does not.

Object motion does not explain that regime. Of the 207 pairs beyond one metre, 167 are ground-truth-static, 39 are annotated as moved, and one is non-rigid. That composition points to view-dependent partial observation: a couch scanned left-half in one session and right-half in the next yields a metre-scale centroid shift under perfect masks and perfect alignment. Fig. 1 therefore shows a centroid-unreliability regime, and appearance is the complementary cue across it whatever the cause. Conversely, the annotation and the measured axis are not co-extensive: 301 of the 383 moved instances sit below

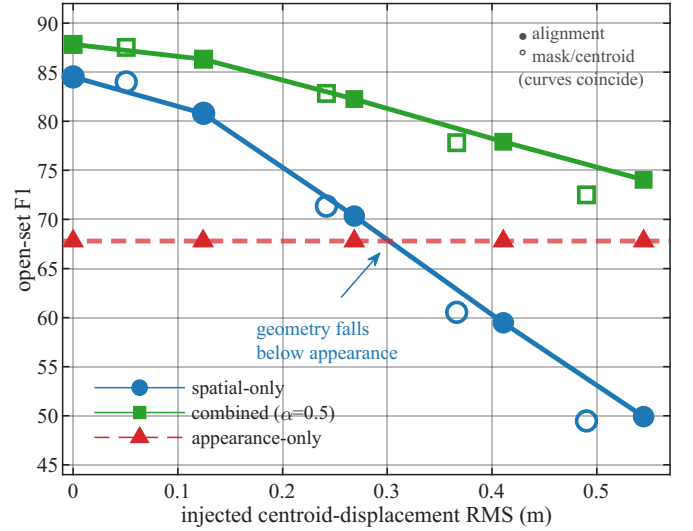


Fig. 2. The centroid prior’s sensitivity tracks the magnitude of centroid error rather than its source. Corrupting the cross-session alignment (filled markers) and jittering per-instance centroids (open markers) nearly coincide when plotted against the *injected* centroid-displacement RMS they add, that is the realized RMS with the ≈ 0.68 m oracle-mask baseline removed in quadrature. Geometry crosses below appearance near 0.30 m. Appearance (DINOv3) does not move, and the fixed combination stays strongest throughout. Means over 20 realizations on the common both-present population, where the spatial oracle scores 84.5.

half a metre.

But that complementarity is specific, and the same-class strata bound it. Stratified by distance to the nearest same-class neighbour and by same-class distractor count, appearance *never* overtakes geometry, and its deficit is worst precisely under the heaviest confusion: the gap is -32.3 when a same-class neighbour sits within half a metre, -26.3 with four or more distractors, and -31.8 inside annotated ambiguity groups, against -8.0 for isolated instances. Frozen appearance rescues the geometry-broken case, not the geometry-ambiguous one. This confirms quantitatively, at the global-embedding tier, a hypothesis OASIS-Map raises qualitatively [12].

The consequence for fusion is that one global weight is not enough. The fixed combination tuned on train is the best single system overall, but beyond a metre the DINOv3 fusion used in these strata is worse than appearance alone (36.3 vs. 59.5; paired $+23.2$ for appearance, 95% CI $[+14.2, +34.1]$), because a weight fitted to the common small-distance regime under-uses appearance exactly where appearance is what works.

C. Two oracles, one scalar

The spatial prior is handed two oracles, the known alignment and the ground-truth masks, and both feed one quantity: the per-instance centroid. We perturb each independently and re-solve, all else fixed, over 20 realizations per level on the common both-present population, where the unperturbed spatial oracle scores 84.5 and the fixed combination 87.8 (against 84.8 and 87.8 on the pooled set of Table I). Alignment is corrupted by SE(3) noise (translation RMS 0.05–0.20 m with co-varying rotation $2\text{--}10^\circ$); masks are modelled by jittering

every centroid on both scans with independent Gaussian noise ($\sigma \in [0.02, 0.20]$ m).

The two perturbations differ in structure, not only in magnitude: an SE(3) error is coherent across every instance in a scan and preserves relative inter-object geometry, whereas centroid jitter is independent per instance and destroys it. Plotted against the injected centroid-displacement RMS each adds, they nonetheless nearly coincide (Fig. 2). Under an absolute-distance rejection rule, then, centroid error RMS is a sufficient statistic for this prior’s sensitivity over the two noise structures tested. Geometry falls to 49.9 F1 as the injected RMS grows to 0.54 m and drops below appearance near 0.30 m, reached by roughly 0.11 m of alignment translation error or $\sigma \approx 0.12$ m of per-instance centroid jitter. Appearance, which never sees the alignment, is essentially unaffected at 67.8; the fixed combination degrades gracefully to 74.0 and stays strongest throughout. Appearance does still consume the instance mask for its crop: replacing oracle masks with raw bounding-box crops changes its F1 from 67.8 to 62.1, so it is the less oracle-dependent cue here rather than an oracle-free one.

Gaussian centroid jitter models the metric component of segmentation error and the SE(3) noise is zero-mean, so each curve is a lower bound on the sensitivity a deployed front-end would see. The headline geometric advantage therefore holds under oracle instance masks and known scan alignment and inverts under moderate centroid error from either source.

IV. IMPLICATIONS FOR LIFELONG SEMANTIC MAPS

Appearance is a conditional cost. Read as a cost model, Table I inverts the usual accuracy/compute story. The most accurate single cue needs no backbone forward pass at all, and every appearance backbone used alone spends 43–91 ms per view (16.9–175.3 GFLOPs) for a net accuracy loss against geometry. Both cues are charged above one shared front-end: each reads the same instance masks, so segmentation cancels in the comparison. The cross-session transform does not cancel; its estimation [25] sits outside this accounting. Among appearance backbones DINOv3 ViT-B is a non-dominated frontier point, 67.8 F1 at 62.7 ms and 17.2 GFLOPs, dominating the heavier ViT-L variants but not the faster base backbones, which buy less accuracy for less latency. The dense patch tier costs more still, storing K token maps per instance instead of one pooled vector and matching them bidirectionally rather than by a single cosine, and it returns its gain only through an explicit confidence heuristic. A lifelong mapper that runs a foundation backbone on every instance in every session is therefore paying continuously for an advantage that materializes in a minority of cases.

Two routes into the same regime. Sections III-B and III-C reach the same boundary from different directions: appearance takes over when the centroid the spatial prior depends on stops being trustworthy, whether because partial views truncate it or because the alignment and mask that produce it are imperfect. Observed error on real rescans is anisotropic, consistent with partial-view truncation, and its crossing sits beyond a metre of measured centroid distance; the injected error of Fig. 2

is isotropic and zero-mean and its crossing sits near 0.30 m of injected RMS, about 0.74 m realized once the 0.68 m the oracles already leave is included. Because the observed and injected errors differ in structure, the two crossings are reported each against its own axis. In both, the quantity that decides the ranking is how wrong the centroid is.

Using the stratification at test time. The measured centroid distance of a *candidate* pair is observable at test time, since it is the spatial prior’s own input; the *correspondence* it belongs to is not. Abstention sidesteps that gap. Escalating only where the geometric matcher declines to claim an instance, with both stages kept at their existing train-tuned thresholds so that no constant is fitted on validation, stays within 1.33 F1 of always-on fusion (95% CI $[-2.25, -0.52]$) while avoiding 81.1% of *rescan*-side backbone passes (the reference side is the map, embedded once offline; 36.9% of it is consulted). The same policy is +1.99 F1 $[+1.07, +3.02]$ over the backbone-free geometric prior, and it recovers only instances geometry declines to claim, so this is a lower bound on regime-aware routing. Budget-matched random routing of identical size, with the same replacement rule, is 22.7 F1 worse, locating the gain in the escalation criterion rather than its budget. Under degraded alignment the first stage abstains more, so 81.1% is an upper bound on the online saving for the appearance branch; the first stage reads the masks and alignment the session already computes.

Scope. These results characterize frozen 2D features on real cross-session rescans of 3RScan under oracle instance masks and known scan alignment, across three representation tiers, with open-set rejection and per-instance cost; automatic front-ends, trained 3D and 4D instance models under their own protocols [10], [26], and the closed-loop integration of a regime-aware policy lie outside this evaluation scope.

The headline for a lifelong semantic map is a boundary rather than a verdict on foundation features. Without a backbone pass, a centroid under known alignment is the strongest cross-session identity cue we measured. Frozen appearance takes over once that centroid becomes unreliable. The changeover between them can be measured.

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